WISH - 2026 - Inductors

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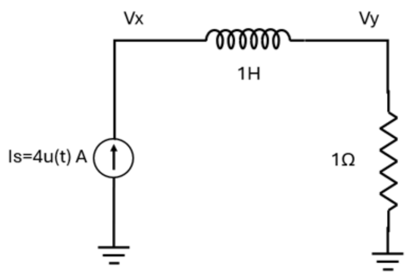
Q1. For the below circuit, the initial current through the inductor is 0A. Find Vx and Vy at time t=5s?

(a) Vx=0V; Vy=0V
(b) Vx=4V; Vy=0V
(c) Vx=0V; Vy=4V
(d) Vx=4V; Vy=4V
(e) Vx=∞V; Vy=0V
(f) Vx=0V; Vy=∞V
(g) Vx=∞V; Vy=∞V
(h) Vx=0V; Vy=1V



Answer the Question 1

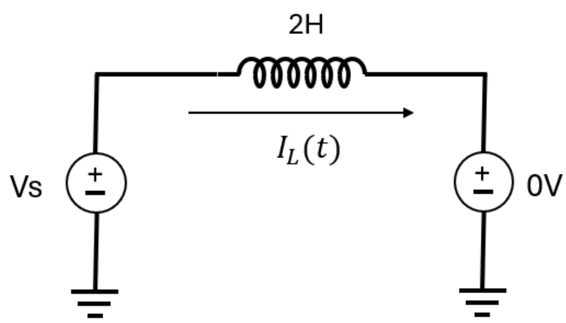
- ○ a
- ○ b
- ○ c
- ○ e
- ○ f
- ○ g

○ d

○ h

Q2. For the below circuit Vs(t)=2u(t)+4u(t-1) and $I_L(t = 0^-)$=0A. What is the current through the inductor at t=5s?

(a) 9A
(b) 10A
(c) 11A
(d) 12A
(e) 13A
(f) 14A
(g) 15A
(h) 16A



Answer the Question 2

○ a

○ b

○ c

○ d

○ e

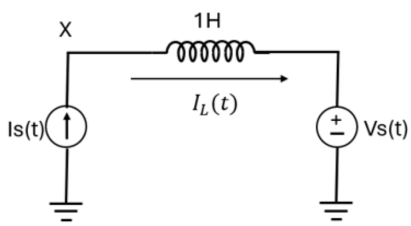
○ f

○ g

○ h

Q 3. For the below circuit Vs(t)=2u(t) V, Is(t)=2r(t) A, and $I_L(t = 0^-)$=0A. What is the voltage at X at 1s?

(a) 0 V
(b) 1 V
(c) 2 V
(d) 3 V
(e) 4 V
(f) 5 V
(g) 6 V
(h) 7 V



Answer the Question 3

○ a

○ b

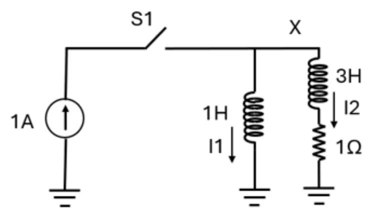
○ e

○ f

- ○ c
- ○ d
- ○ g
- ○ h

Q 4. For the below circuit the initial currents through the inductors are 0A. S1 is closed at t=0s. Find I1 at $t=0^+$ and $t=\infty$?

(a) $I1(t=0^+)=\frac{3}{4}A$; $I1(\infty)=1A$
(b) $I1(t=0^+)=\frac{1}{4}A$; $I1(\infty)=0A$
(c) $I1(t=0^+)=1A$; $I1(\infty)=1A$
(d) $I1(t=0^+)=0A$; $I1(\infty)=0A$
(e) $I1(t=0^+)=\frac{1}{2}A$; $I1(\infty)=\frac{1}{2}A$
(f) $I1(t=0^+)=\infty A$; $I1(\infty)=0A$
(g) $I1(t=0^+)=0A$; $I1(\infty)=\infty A$
(h) $I1(t=0^+)=\infty A$; $I1(\infty)=\infty A$



Answer the Question 4

- ○ a
- ○ b
- ○ c
- ○ d
- ○ e
- ○ f
- ○ g
- ○ h

Q5. In question 4, the highest instantaneous voltage at node X is

(a) 0V
(b) ∞V
(c) 1V
(d) 10V
(e) -∞V
(f) -1V
(g) -10V
(h) -4V

Answer the Question 5

○ a

○ b

○ c

○ d

○ e

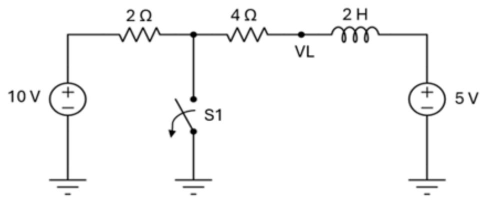
○ f

○ g

○ h

Q6. The circuit below has switch S1 in closed state for a very long time. At t=0, S1 is suddenly opened. What is the voltage at node VL at t = 0^+ ?

(a) -12.5 V
(b) -5 V
(c) -3.33 V
(d) 0
(e) 2.5 V
(f) 6.66 V
(g) 10 V
(h) 17.5 V



Answer the Question 6

○ a

○ b

○ c

○ d

○ e

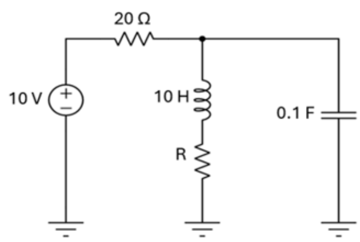
○ f

○ g

○ h

Q7. What is the value of R in the circuit below such that energies stored in the inductor and capacitor are equal at t = ∞?

(a) 1 Ω
(b) 2.5 Ω
(c) 4 Ω
(d) 8.66 Ω
(e) 10 Ω
(f) 32 Ω
(g) 50 Ω
(h) 100 Ω

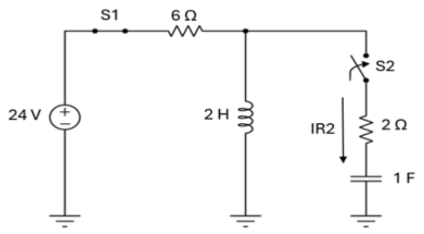


Answer the Question 7

- ○ a
- ○ b
- ○ c
- ○ d
- ○ e
- ○ f
- ○ g
- ○ h

Q8. The circuit below had switch S1 in closed state and switch S2 in open state for a very long time. At t = 0, S2 is suddenly closed while S1 remains closed throughout. Assuming capacitor to be initially uncharged, what is the current IR2 at $t = 0^+$?

(a) -4 A
(b) -3 A
(c) -1 A
(d) 0 A
(e) 1 A
(f) 3 A
(g) 4 A
(h) 12 A



Answer the Question 8

- ○ a
- ○ b
- ○ c
- ○ d
- ○ e
- ○ f
- ○ g
- ○ h

Q9. In question 3, what is the current through IR2 at t = 0^{+} if the capacitor is initially charged to +4V before switch S2 is closed?

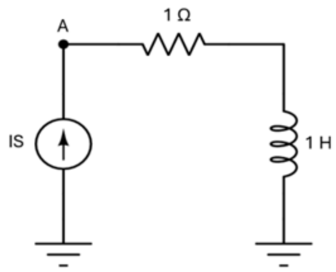
(a) -3.5 A
(b) -2 A
(c) -0.5 A
(d) 0 A
(e) 0.5 A
(f) 1.5 A
(g) 3 A
(h) 5 A

Answer the Question 9

- a
- b
- c
- d
- e
- f
- g
- h

Question 10-> IS = 2r(t). Voltage at node A at t = 5s is:

(a) -5 V
(b) -3.2 V
(c) -0.5 V
(d) 0 V
(e) 3 V
(f) 8 V
(g) 10 V
(h) 12 V



Answer the Question 10

- a
- b
- c
- d
- e
- f
- g
- h

Done